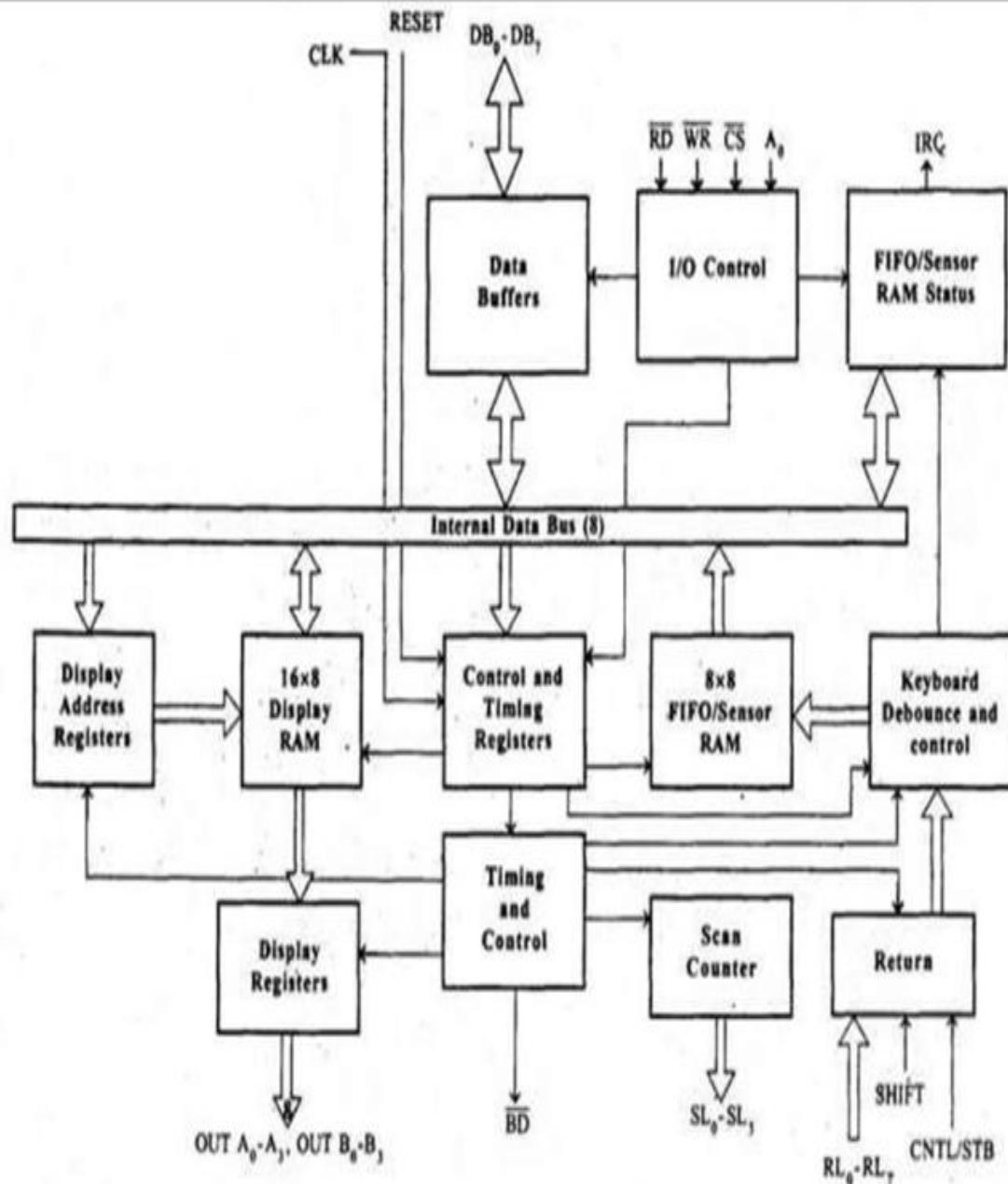


Architecture of 8279 programmable keyboard/display controller



1. I/O Control and Data Buffer
2. Control and Timing Register and Timing Control
3. Scan Counter
4. decoded scan mode
5. Return Buffers, Keyboard Debounce, and Control
6. FIFO/Sensor RAM and Status Logic
7. Display Address Registers and Display RAM

- The 8279 keyboard can handle **up to 64 keys** and sends key codes to the CPU.
- It **removes noise (debouncing)** and stores key data in an **8-byte FIFO memory**.
- If more than **8 keys are pressed**, an **overflow error** occurs.
- When a key is pressed, the CPU can read it by:
 - **Interrupt** (automatic signal), or
 - **Polling** (CPU checks manually)
- After reading, the key is **removed from FIFO**, making space for new keys.
- **How Many Ways the Keyboard is Interfaced with the CPU?**
- The Keyboard can be interfaced either in the interrupt or the polled mode. In the **Interrupt mode**, the processor is requested service only if any key is pressed, otherwise the CPU will continue with its main task.
- In the **Polled mode**, the CPU periodically reads an internal flag of 8279 to check whether any key is pressed or not with key pressure.

Operational Modes of 8279

There are two modes of operation on 8279 – **Input Mode** and **Output Mode**.

Input Mode

This mode deals with the input given by the keyboard and this mode is further classified into 3 modes. 1. Scanned Keyboard Mode, 2. Scanned Sensor Matrix, 3. Strobed Input

Output Mode

This mode deals with display-related operations. This mode is further classified into two output modes. 1. Display Scan, 2. Display Entry

Interfacing a Microprocessor To Keyboard

- When you press a key on your computer, you are activating a switch.
There are many different ways of making these switches.
 1. Mechanical key switches
 2. Membrane key switches
 3. Capacitive key switches
 4. Hall effect key switches

Mechanical key switches

- In mechanical-switch keys, two pieces of metal are pushed together when you press the key.

Membrane key switches

- These switches are really a special type of mechanical switches. They consist of a three-layer plastic or rubber sandwich.

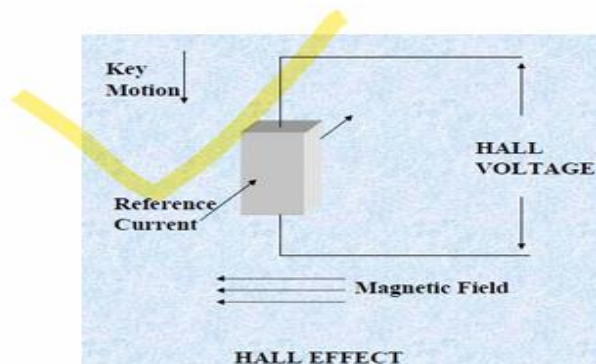
Capacitive key switches:

- A capacitive key switch has two small metal plates on the printed circuit board and another metal plate on the bottom of a piece of foam.

Hall effect key switches

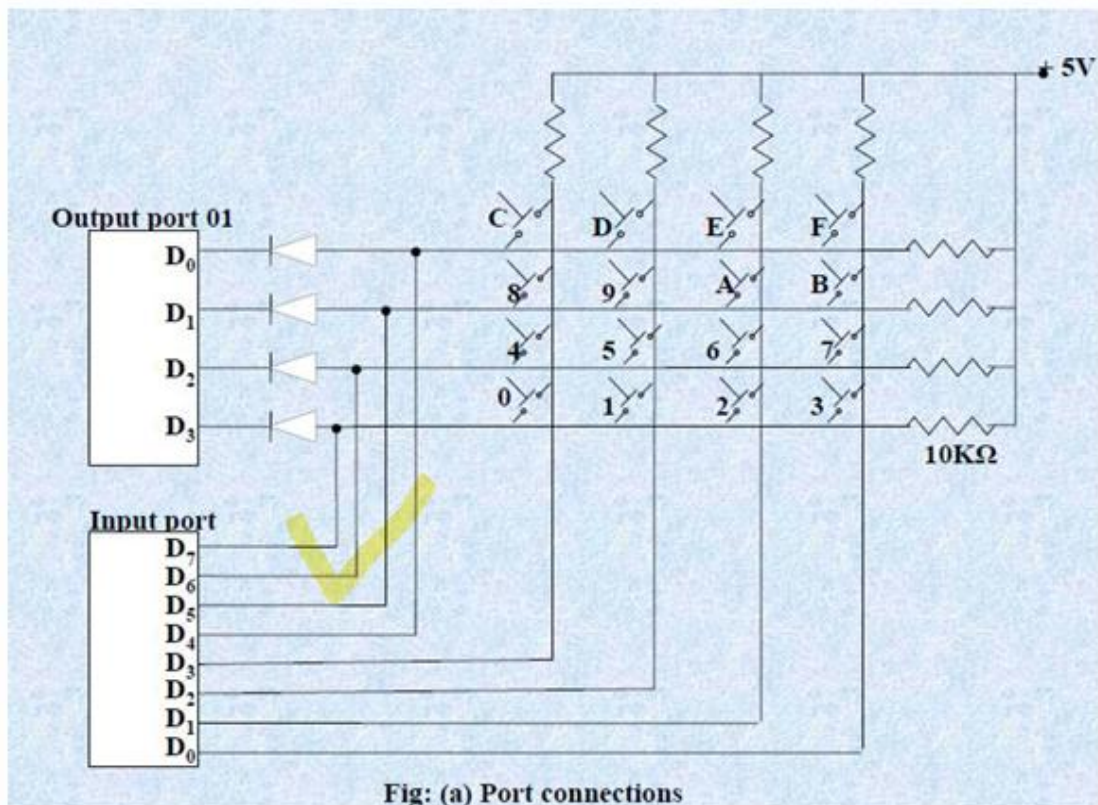
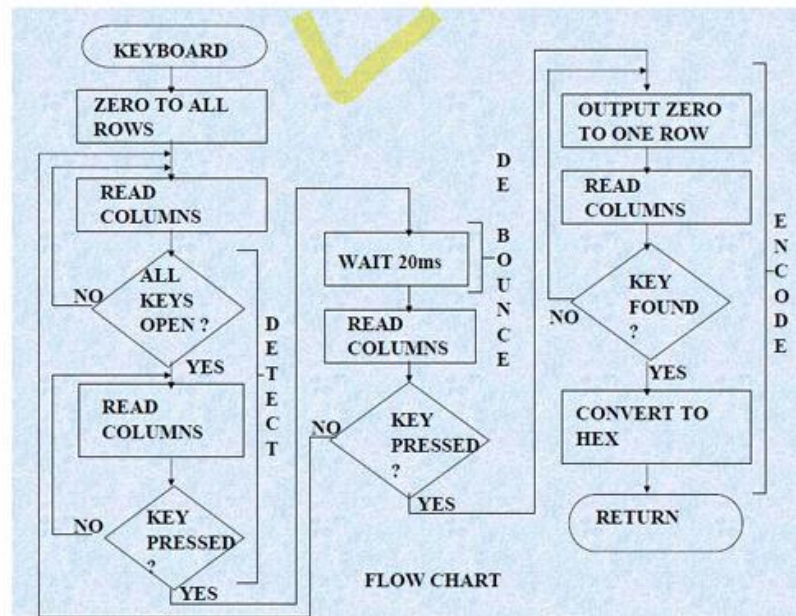
- This is another type of switch which has no mechanical contact.

Hall effect key switches

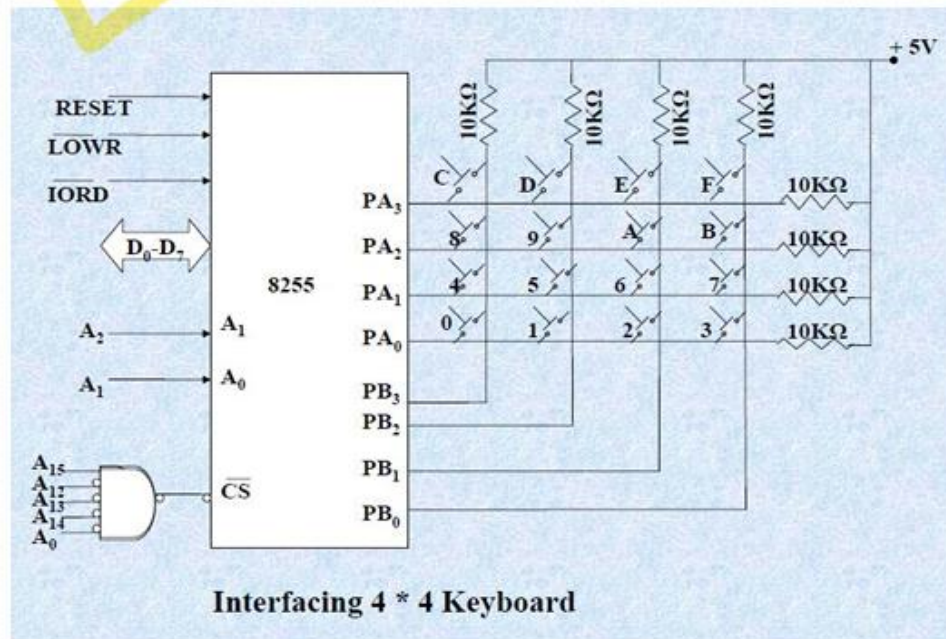


- Getting meaningful data from a keyboard, it requires the following three major tasks:
 1. Detect a keypress
 2. Debounce the keypress
 3. Encode the keypress

Software Keyboard Interfacing



Keyboard Interfacing with Hardware



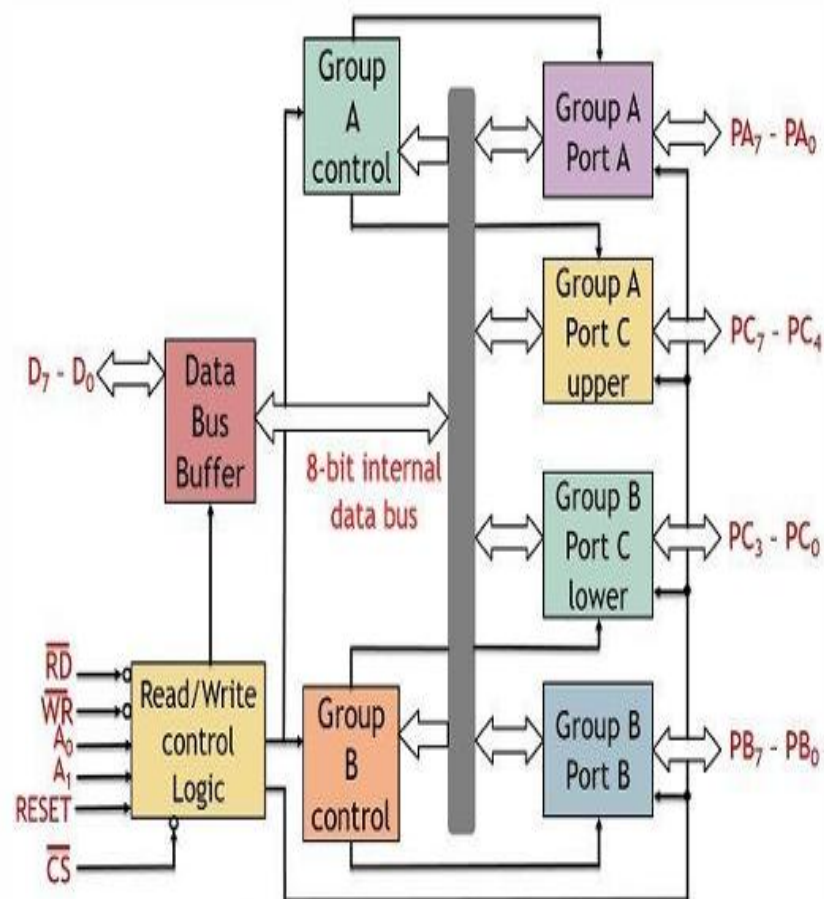
- The higher order lines of port A and port B are left unused.
- The address of port A and port B will respectively 8000H and 8002H while address of CWR will be 8006H.
- The control word for this problem will be 82H.
- Code segment CS is used for storing the program code.
- **Key Debounce** : Whenever a mechanical push-button is pressed or released once, the mechanical components of the key do not change the position smoothly, rather it generates a transient response .

Example

- Interface a 4 * 4 keyboard with 8086 using 8255 and write an ALP for detecting a key closure and return the key code in AL. The debounce period for a key is 10ms. Use software debouncing technique. DEBOUNCE is an available 10ms delay routine.

- **Solution:** Port A is used as output port for selecting a row of keys while Port B is used as an input port for sensing a closed key.
- Thus the keyboard lines are selected one by one through port A and the port B lines are polled continuously till a key closure is sensed.
- The routine DEBOUNCE is called for key debouncing.
- The key code is depending upon the selected row and a low sensed column.

Architectural representation of 8255 PPI:



Architecture of 8255 PPI

- The two modes in which 8255 can be programmed are as follows:
 - Bit set/reset mode
 - I/O mode

The bits of port C gets set or reset in the BSR mode.

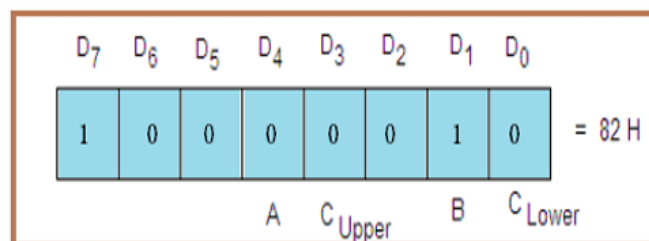
The other mode of 8255 i.e., I/O mode is further classified into:

Mode 0: Simple input/output

Mode 1: Input output with handshaking

Mode 2: Bidirectional I/O handshaking

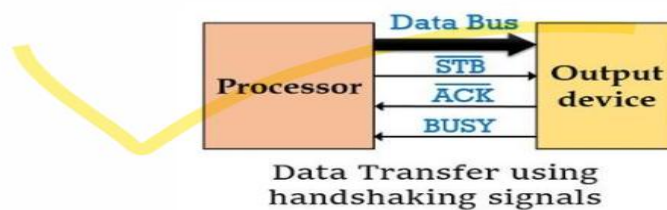
- **Mode 0:** Input/Output mode
- This mode is the simple input output mode of 8255 which allows the programming of each port as either input or output port. The input/output feature of mode 0 includes:
 - Mode 1 and Mode 2 both are same but the only difference is mode 1 does not support bidirectional handshaking.
- Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port.
- **Solution.** The control word for this case shown in bellow.



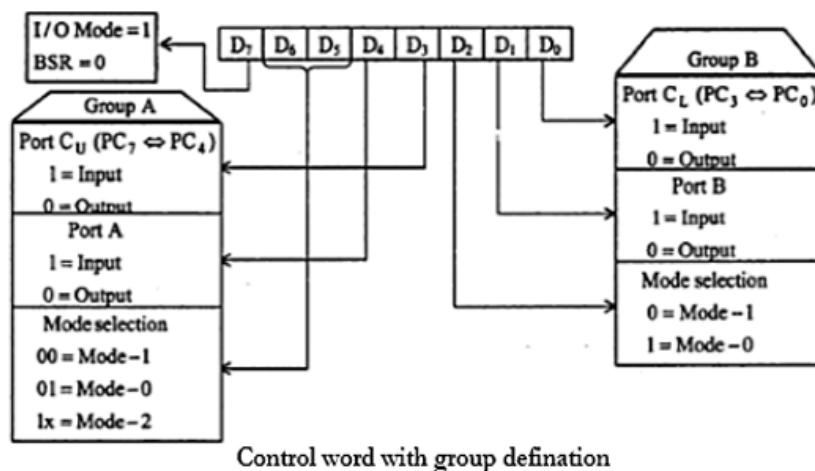
The table below shows the operation of the control signals:

A ₁	A ₀	RD'	WR'	CS'	Input/Output Operation
0	0	0	1	0	Port A - Data Bus
0	1	0	1	0	Port B - Data Bus
1	0	0	1	0	Port C - Data Bus
0	0	1	0	0	Data Bus - Port A
0	1	1	0	0	Data Bus - Port B
1	0	1	0	0	Data Bus - Port C
1	1	1	0	0	Data Bus - Control register

The figure below shows the data transferring between CPU and an output device having different operating speeds:



The control word format of the 8255 is shown in Fig. below.



- There are 16 combinations of control words for various configurations of the ports of 8255 for Mode 0 operations. These control words are shown in table (next slide).

Table

Control Word Bits								Control Word	PORT A	PORT C _{Upper}	PORT B	PORT C _{Lower}
D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀					
1	0	0	0	0	0	0	0	80 H	Output	Output	Output	Output
1	0	0	0	0	0	0	1	81 H	Output	Output	Output	Input
1	0	0	0	0	0	1	0	82 H	Output	Output	Input	Output
1	0	0	0	0	0	1	1	83 H	Output	Output	Input	Input
1	0	0	0	1	0	0	0	88 H	Output	Input	Output	Output
1	0	0	0	1	0	0	1	89 H	Output	Input	Output	Input
1	0	0	0	1	0	1	0	8A H	Output	Input	Input	Output
1	0	0	0	1	0	1	1	8B H	Output	Input	Input	Input
1	0	0	1	0	0	0	0	90 H	Input	Output	Output	Output
1	0	0	1	0	0	0	1	91 H	Input	Output	Output	Input
1	0	0	1	0	0	1	0	92 H	Input	Output	Input	Output
1	0	0	1	0	0	1	1	93 H	Input	Output	Input	Input
1	0	0	1	1	0	0	0	98 H	Input	Input	Output	Output
1	0	0	1	1	0	0	1	99 H	Input	Input	Output	Input
1	0	0	1	1	0	1	0	9A H	Input	Input	Input	Output
1	0	0	1	1	0	1	1	9B H	Input	Input	Input	Input

How can we assign input and output port of 8255?

Introduction to Printers

- Printers are commonly used output devices that produce a hard copy of document stored in electronic form, i.e they put information from computer on to paper.
- There are various kinds of printers available today like Impact printers, Bubble-jet printers, Laser printers, Thermal printers etc.
- **Types of printers**
 - Impact
 - Solid-Ink Printers
 - Thermal Printers
 - Inkjet Printers
 - Laser printers

Introduction to Scanners

Scanner is a device that scans images, printed text, and handwriting etc. and converts it to digital form or image.

Components inside a scanner:

- Glass Plate and Cover • Scanning head • Stepper Motor

Types • Flatbed Scanners • Sheetfed Scanners • Handheld Scanners

There are eight major communication types

1. Serial Communication
2. Parallel Communication
3. USB (Universal Serial Bus)
4. Network (Ethernet)
5. Infrared Communication
6. SCSI (Small Computer System Interface)
7. IEEE 1394 (FireWire)
8. Wireless Communication

Printers And Scanners Problems And Troubleshooting

- Laser printer problems
 - Print results too light, white/light columns
 - Ghosting
 - Prints black pages
 - Memory Overflow error
 - Paper jams
 - Repetitive small marks or defects
 - Image Smudging

Printers And Scanners Problems And Troubleshooting

- Dot-Matrix printer problems
 - Print results are too light
 - Printer does not print
 - Characters are incomplete
 - Printout jams inside the printer
 - Stepper motor problems

Scanner Problems and Troubleshooting

- Common problems with scanner are scanners won't turn on, strange noises from scanner, Scanners won't scan etc.
- Tips to troubleshoot the basic scanner related problems are:
 - Ensure that the scanner software is properly installed and all the settings are properly configured
 - Make sure that the power light on the scanner is turned on.
 - If any software related problems are noticed in scanner then uninstall and reinstall the scanner software
 - Try to perform the basic tests that are provided by the manufacturer.
 - Make sure that the cables to the scanner are properly connected.
 - Overheating, dust and humid environment can cause permanent damage to the scanner.
 - Restarting your computer, unplugging cables and verifying any hardware conflicts in the device manager can diagnose and

Tools To Troubleshooting Problems

- Multimeter
- Screw driver
- Cleaning solutions
- Extension Magnets
- Test patterns

Universal Serial Bus (USB)

- a set of interface specifications for high speed wired communication between electronics systems peripherals and devices with or without PC/computer.

USB Functionality

1. Power Delivery
2. Battery Charging 1.2
3. USB On-the-Go
4. DisplayPort Alt Mode
5. USB Docking.

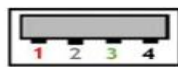
Types of USB Ports and Connectors



USB Connector Pinouts

Pin No:	Signal	Color of the cable
1	+5V power	Red
2	- Data	White / Yellow
3	+Data	Green / Blue
4	Ground	Black/Brown

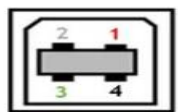
USB Type A and Type B pinout



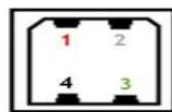
Female



Male



Female

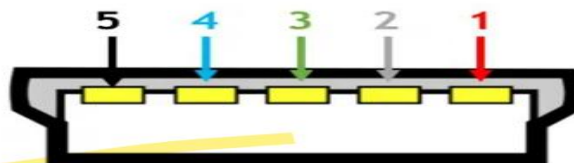


Male

1 Vcc	Red	+5V
2 D-	White	Data-
3 D+	Green	Data+
4 GND	Black	Ground

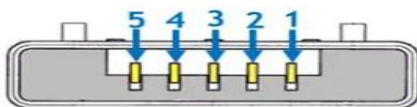
www.Technonhiles.com

Pinout of Mini B

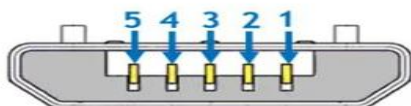


1 Vcc	Red	+5V
2 D-	White	Data-
3 D+	Green	Data+
4 ID	N/A	USB OTG ID
5 GND	Black	Ground

Micro A & Micro B pinout



Pinout of Micro A



Pinout of Micro B

1 Vcc	Red	+5V
2 D-	White	Data-
3 D+	Green	Data+
4 ID	N/A	USB OTG ID
5 GND	Black	Ground

- **Features of USB standard 3**
- The Standard 3 models are accepted globally due to the following benefits:
 1. It is capable to transfer data with a speed of 5Gbps and above.
 2. It has larger bandwidth.
 3. It offers better power management.

4. It improves the bus usage.
5. It includes additional ground and dedicated pins for transmission and reception

What's New in USB4?

1. Up to 40Gbps Data Transfer Rate
2. Multiple Data and Display Protocols
3. Video and Data Bandwidth Optimization
4. 100Watt Charging
5. Backward compatibility with USB 3.2, USB 2.0 and Thunderbolt 3.

The Host controllers

the host controller and the root hub are part of the computer hardware. Types of Host Controller:

1. OHCI (Open Host Controller Interface)
2. UHCI (Universal Host Controller Interface)
3. EHCI (Extended Host Controller Interface)
4. VHCI (Virtual Host Controller Interface)
5. USB4 Host Interface

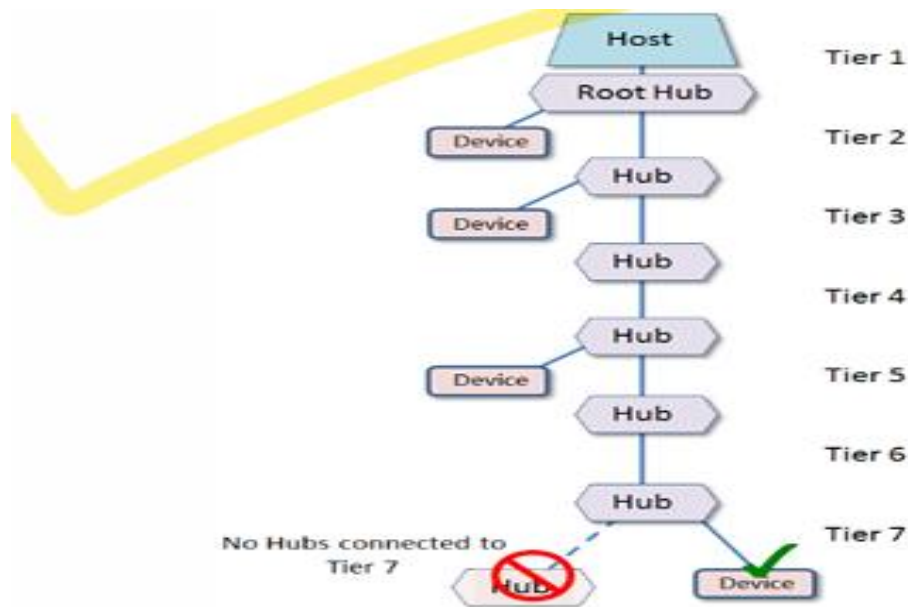
- **USB 4 vs. Thunderbolt, Which Is Better?**

Feature	USB4	Thunderbolt 4
Max Speed	Up to 40 Gbps	40 Gbps (guaranteed)
Display	Optional	2× 4K required
Daisy Chain	Optional	Up to 6 devices
Power	7.5W minimum	15W minimum
Flexibility	High	Strict standard

Thunderbolt 4 is “better” in reliability and guaranteed features.

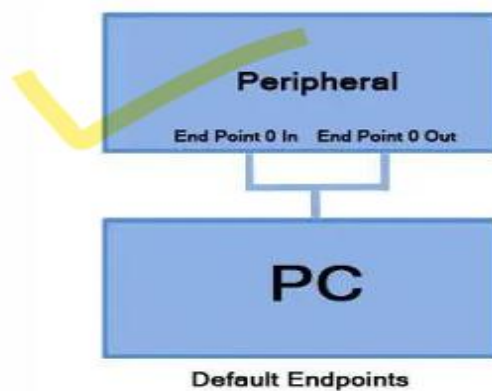
Architecture

- USB uses a tiered-star topology.
- At the center of the star is the *USB host*, which defines



Endpoints

The endpoints are source or sink of information in a communication channel.



Pipes

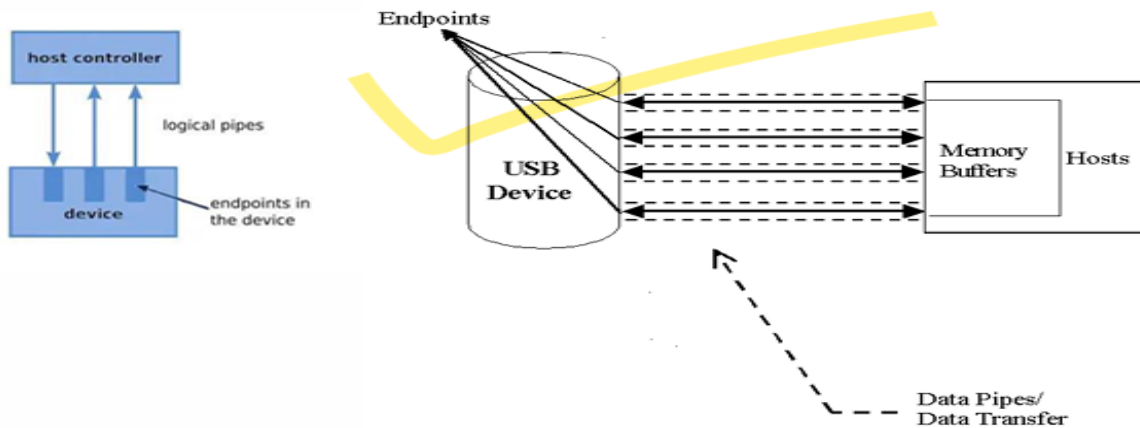
A pipe is a logical data connection between host controller's software and device endpoint. There are two types of pipes:

1. Message pipes: These are bi-directional pipes.

2. Stream pipes: These are unidirectional pipes.

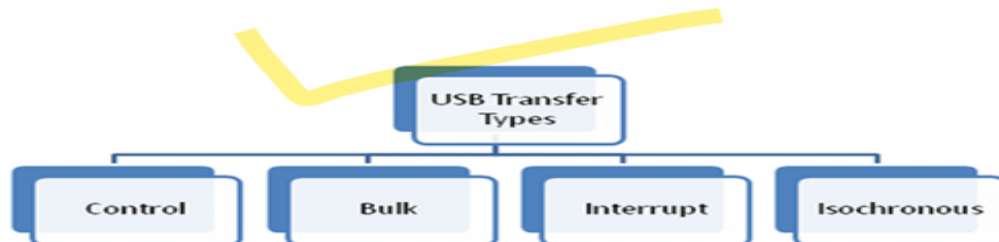
The Default Control pipe is a special type of message pipe which is bidirectional and supports control transfer type.

Pipes



Transfer Types

- There are four types of transfer modes or types which can be used for communication:

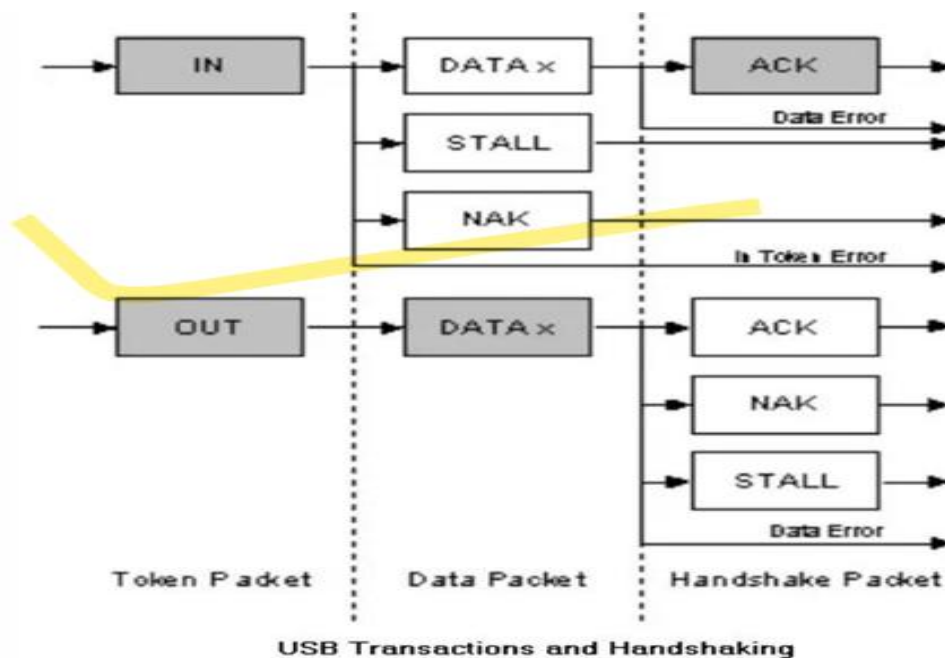


Transaction

- A single transaction contains transmission of up to three packets. These packets are:
 - **Token Packet:** This packet is always sent by Host (Header defining what it expects to follow)
 - **Data Packet:** This packet can be sent by Host or Device. (Containing the payload)
 - **Handshake / Status Packet:** This packet provides success/failure information for the data packet received. If Host transmits the data packet, the device sends the

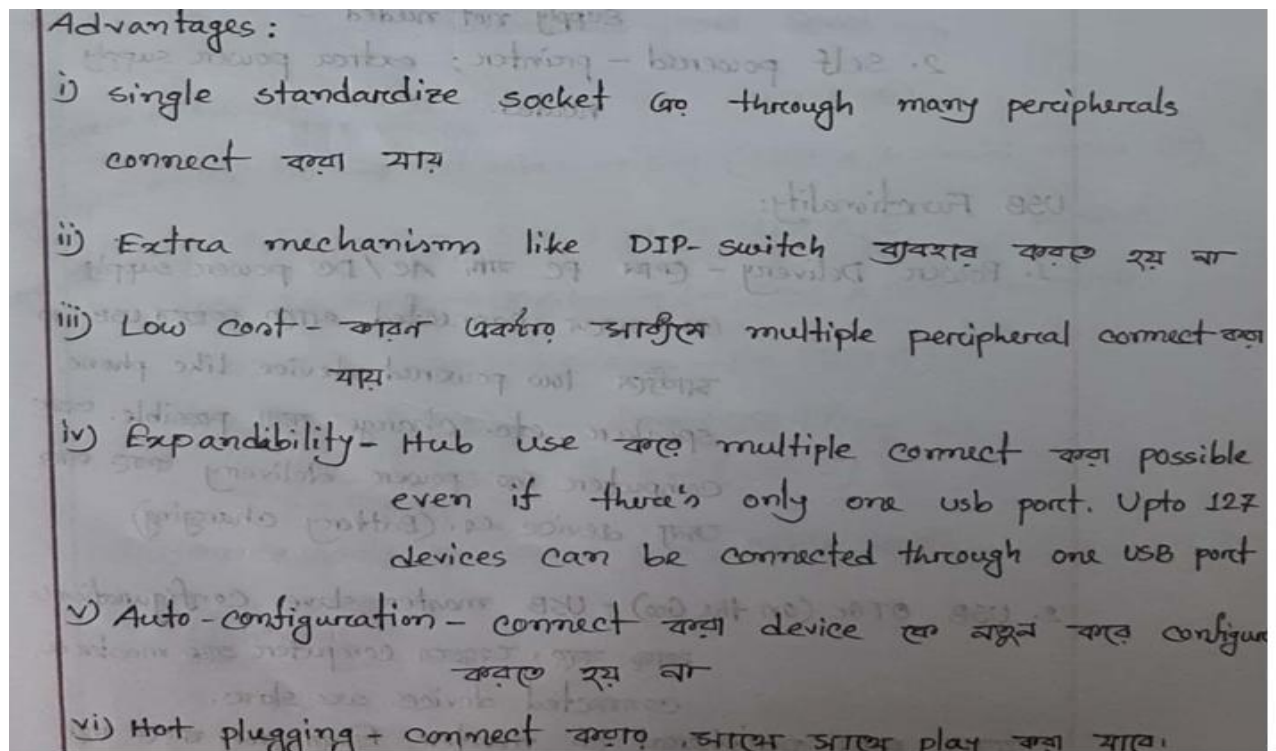
Handshaking

- Handshaking is a mechanism to check the success/failure of a request or to check the delivery of a packet.
- It is done to avoid loss of packets and to ensure successful transmission.
- Terms related to handshaking:
 - **ACK** – acknowledgment for data receive (success)
 - **NAK** – negative acknowledgment means no data
 - **STALL** – request not supported or endpoint halted
 - **NYET** – not yet. A case can be when device is busy and not ready for next data packet
 - **ERR** – split transaction error
 - **No response**



* Limitations of USB:

- i) Speed: lower than Gigabit Ethernet; Firewire 800 is a competitor of USB.
- ii) Peer-to-peer Communication: is not possible
- iii) Distance: 5 meters or USB hub needed
- iv) Broadcasting: doesn't provide broadcasting feature.

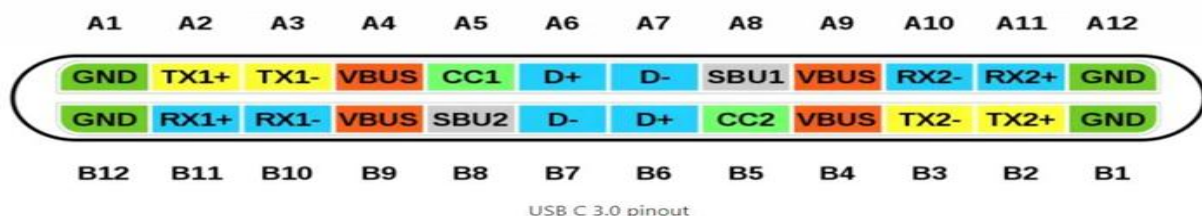


How do the USB peripherals communicate with microprocessor?

USB peripherals communicate with the microprocessor through a USB controller. The microprocessor acts as the host, controls all communication, and requests data from the device. Data is transferred serially in packets, and the USB controller manages the process and notifies the processor when needed.

USB Type C 3.0

- Type C has been the game-changer among all the USB types. It supports data transmission as well as power delivery. And thus, it has become a universal connector for modern devices.



- a. What do you mean by true color representation? Does it represent any disadvantages?
How to handle the problems?
- b. If we use 5 bits each for red and blue and 6 bits for green in direct coding method, how many possible simultaneous colors do we have?

True color representation means representing each pixel using 24 bits (3 bytes):

- 8 bits for Red
- 8 bits for Green
- 8 bits for Blue

So, each color channel has 256 levels → total colors = 16,777,216 colors.

👉 This gives very realistic, natural images (like photos).

bit 1: <i>r</i>	bit 2: <i>g</i>	bit 3: <i>b</i>	color name
0	0	0	black
0	0	1	blue
0	1	0	green
0	1	1	cyan
1	0	0	red
1	0	1	magenta
1	1	0	yellow
1	1	1	white

Fig. 2-3 Direct coding of colors using 3 bits.

true color representation intensity

✗ Disadvantages of True Color

1. **High memory usage**
Each pixel needs 24 bits → large image size
2. **High bandwidth required**
More data must be transferred
3. **Slow processing**
More data → more computation
4. **High cost**
Needs better hardware (graphics card, memory)

✓ How to Handle True Color Problems

- Use image compression (JPEG, PNG)
- Use fast data transfer (Buffer, DMA)
- Use GPU for faster processing
- Reduce color depth (16-bit, 8-bit)
- Reduce image resolution

(b) Number of colors (5 bits Red, 6 bits Green, 5 bits Blue)

Here:

- Red = 5 bits $\rightarrow 2^5 = 32$
 - Green = 6 bits $\rightarrow 2^6 = 64$
 - Blue = 5 bits $\rightarrow 2^5 = 32$
- } Levels

Total colors:

$$\text{Total Colors} = 2^5 \times 2^6 \times 2^5 = 2^{16} = 65536$$

👉 Answer: 65,536 colors



Peripherals

- Peripherals are any device that can be attached to a CPU, e.g. hard disk, mouse, printer.

Interfaces

- What is an interface?
 - The interface is the combination of hardware and software needed to link the CPU to the peripherals and to enable them to communicate

Example: USB, Printers

- The main functions of an interface that you need to know about are:
 - Buffering;
 - Converting data to and from serial and parallel forms;
 - Converting data to and from analogue and digital forms;
 - Voltage conversion;
 - Protocol conversion; &
 - Handling of status signals.
- Computer peripherals have different characteristics. For example, they may:
 - Have different data transfer rates;
 - Use a wide variety of codes and control signals;
 - Transmit data in serial or in parallel form;
 - Even work at higher voltages than the CPU; &
 - All operate at much slower speeds than the CPU.

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5

What is a Buffer?

- A buffer is an area of memory used for the transfer of data between a computer and a peripheral.
- A buffer provides temporary storage of data.

Why use buffers?

- Peripherals operate at much slower speeds than the CPU. Using buffers helps the computer system compensate for the differences in operating speeds between CPU and its peripherals.

Spoolers

- Spooling is another technique used in the transfer of data to a slow peripherals. In this case the data

Buffers and Spoolers

- **Buffer v's Spooler**

- If the CPU is very busy and doesn't get much idle time then spooling can be a very slow process.

- **Spooler v's Buffer**

- A buffer is limited by the amount of RAM whereas a spooler uses backing storage which has a very large capacity.

- Use BOTH for optimum efficiency.

Wireless Communication

- Current trends in wireless communication include the standards WiFi and Bluetooth.

Bluetooth

- Bluetooth can make short-range links between personal devices, such as mobile phones and headsets, palmtops (PDA) and laptop (notebook) computers.
- Bluetooth is also used for wireless keyboards and mouse.
- It is expected that later versions of Bluetooth will be able to transmit data at a speed of 2 Mbps and for longer distances

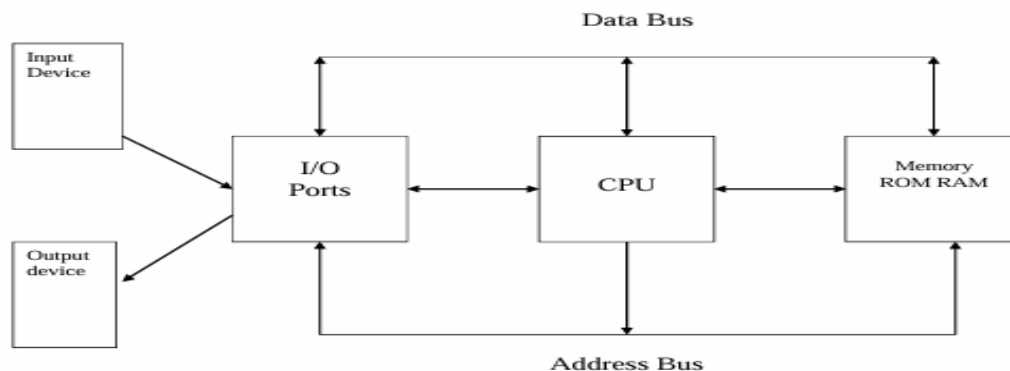
WiFi

- WiFi stands for the **Wireless Fidelity Alliance**.
- The main use for WiFi is in wireless local area networking (WLAN). WiFi devices have typical ranges from 15 to 50 meters and typical data transfer rates from 5 to 20 Mbps.

Types of Computers

- Mainframe Example: IBM 4381
- Super Computer Example: Cray Y-MP/ 832
- Mini Computer Example: IBM 4381
- Micro Computer Example: Desktop, Laptop

Overview of Microcomputer Structure and Operation



Q. interpret EFE16 as unsigned, signed and floating point numbers that micro-processor can understand. Floating point number = Floating point rules follow:

◆ 1. Unsigned Value

$$\text{EFE16}_{16} = 982,550_{10}$$

✓ Unsigned = 982,550

◆ 2. Signed Value (2's Complement)**

👉 Binary MSB = 1 → negative number

$$\begin{aligned}\text{Signed} &= -(2^{20} - 982,550) \\ &= -(1,048,576 - 982,550) = -66,026\end{aligned}$$

• Convert 2.625 to our 8-bit floating point format.

1. The integral part is easy, $2_{10} = 10_2$. For the fractional part:

0.625	$\times 2 =$	1.25	1	Generate 1 and continue with the rest.
0.25	$\times 2 =$	0.5	0	Generate 0 and continue.
0.5	$\times 2 =$	1.0	1	Generate 1 and nothing remains.

So $0.625_{10} = 0.101_2$, and $2.625_{10} = 10.101_2$.

2. Add an exponent part: $10.101_2 = 10.101_2 \times 2^0$.

3. Normalize: $10.101_2 \times 2^0 = 1.0101_2 \times 2^1$.

4. Mantissa: 0101

5. Exponent: $1 + 3 = 4 = 100_2$.

6. Sign bit is 0.

The result is

0	100	0101
---	-----	------

Represented as hex, that is 45_{16} .

* Convert -4.75 to 8-bit floating point format.

$$\Rightarrow (4)_{10} = (100)_2$$

$$0.75 \times 2 = 1.50 \quad 1$$

$$0.50 \times 2 = 1.00 \quad 1$$

$$\therefore (4.75)_{10} = (100.11)_2$$

Normalize: 100.11×2^0 → appended w/o any change in value
 10.011×2^1

1 bit left from ←
 (.) after normalize
 1.0011×2^2 → exponent of 2
 Mantissa = 0011

$$\text{exponent} = 2 + 3 = 5 = 101_2$$

$$\text{sign} = 1$$

$$(-4.75)_{10} = 1 \ 101 \ 0011 = (D3)_{16}$$

sign exponent mantissa

bias k - exponent

k = no of bits in exponent field

$$2^{(k-1)} - 1 = 2^{3-1} - 1 = 2$$

8 bit $k=3$

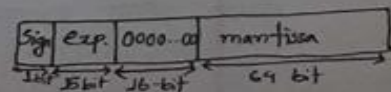
16 " " $k=5$

32 " " $k=8$

64 " " $k=11$

96 " " $k=15$

96 bit $k=15$

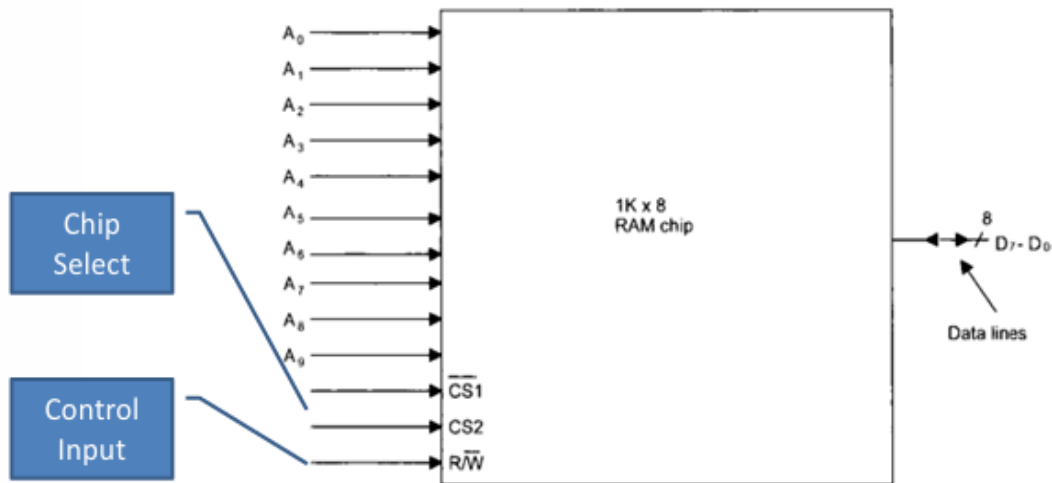


Main Memory Array Design

Memory Array Design means: Interconnecting several memory chips.

A 16 bit microprocessor can address directly a maximum of $2^{16} = 65,536$ or 64K bytes of memory locations.

Main Memory Array Design



Typical 1K × 8 SRAM chip.

TABLE			Truth Table for Controlling SRAM.
$\overline{CS1}$	CS2	R/ $\overline{W}$	Function
0	1	0	Write Operation
0	1	1	Read Operation
1	X	X	The chip is not selected
X	0	X	The chip is not selected

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Main Memory Array Design

- To connect a microprocessor to ROM/RAM chips, three address-decoding techniques are commonly used:
 - **Linear decoding**
 - **Full/Partial decoding**
 - **Memory decoding using PALs**

2. (i) **Partial Address Decoding**: not all address space is implemented; a subset of the address lines are needed to point the physical memory locations.

(ii) **Full Address Decoding**: All the address lines are used to specify a memory location; all addresses are implemented.

• **Linear decoding**

Suppose we have

4K SRAM chip array comprised of the four 1K SRAM chips

➤ This system can be expanded up to a total capacity of 6K using A_{14} , and A_{15} , as chip selects for two more 1K SRAM chips.

Address Range (Hex)	SRAM Chip Number
3800-3BFF	I
3400-37FF	II
2C00-2FFF	III
1C00-1FFF	IV

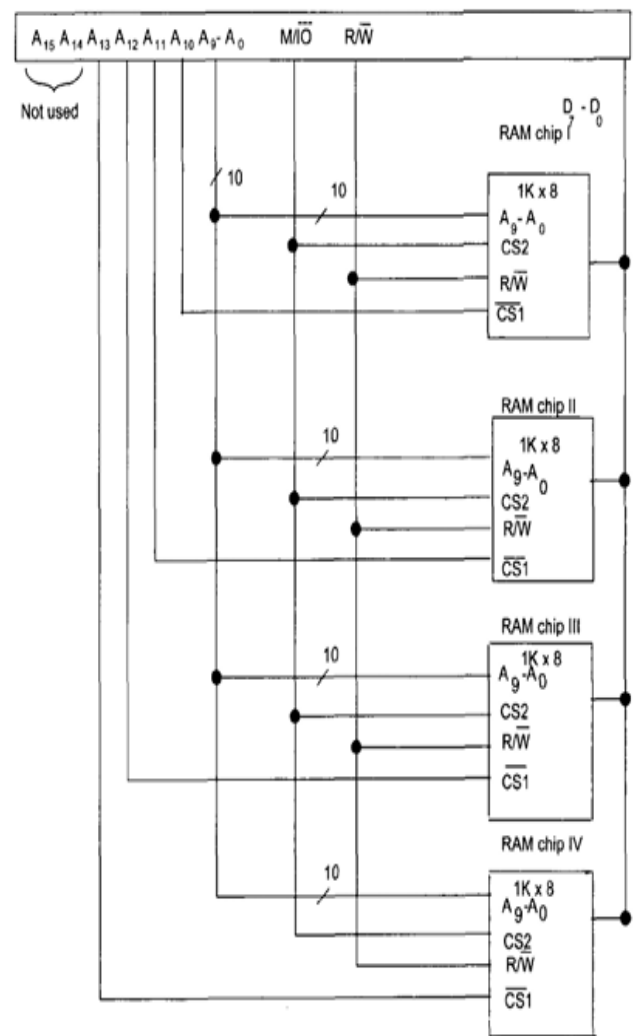


FIGURE 3.8 Microprocessor connected to 4K SRAM using the linear select decoding technique.

- **Linear decoding Advantage**

does not require decoding hardware.

- **Linear decoding Disadvantage**

1. Bus Conflict: two or more of lines A_{10} - A_{13} are low at the same time, more than one SRAM chip are selected.

Solution: software must be written such that it never reads into or writes from any address in which more than one of bits A_{10} - A_{13} are low.

2. Memory Foldback : Wastes a large amount of address space. For example, whenever the address value is B800 or 3800, **SRAM chip I** is selected.

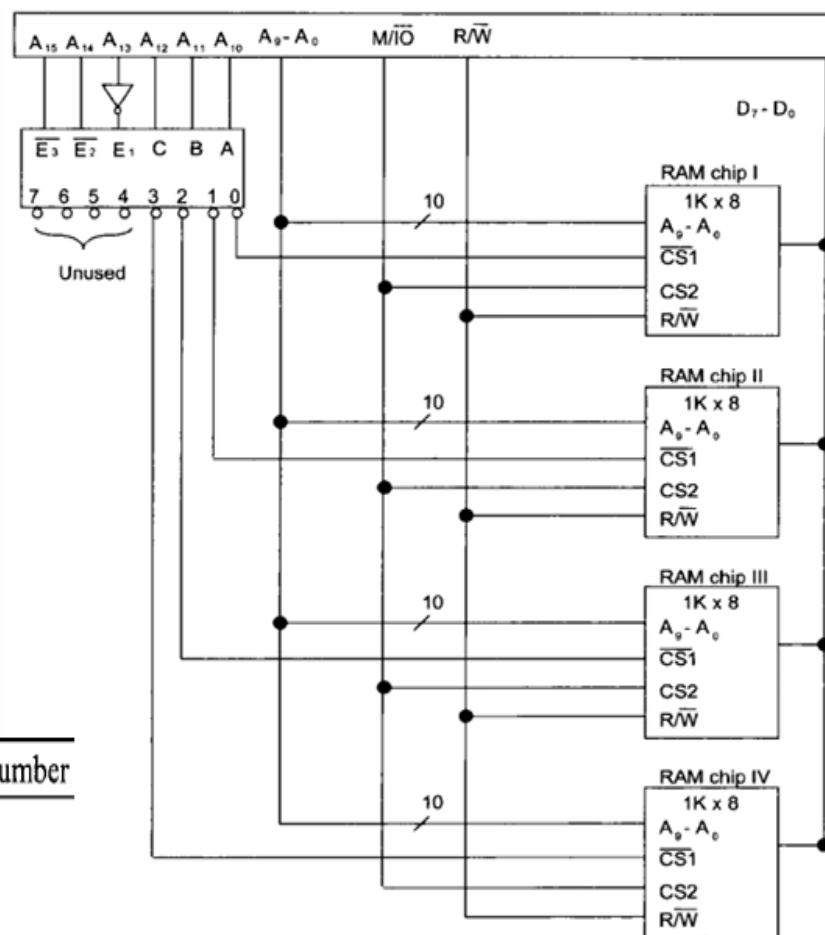
Solution: use full decoded memory addressing.

Full Decoding.

Using 3x8 decoder output selects one of the four 1K SRAM chips, depending on the values of A_{12} , A_{11} , and A_{10}

Decoder output enabled only when $E3 = E2 = 0$ and $E1 = 1$

A_{11}	A_{10}	SRAM Chip Number
0	0	I
0	1	II
1	0	III
1	1	IV



Interconnecting a microprocessor with a 4K RAM using full decoded memory addressing.

Example: 01

- A processor having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor.
 1. What is the memory capacity of the processor?
 2. How many addresses are required for a RAM?/
 3. How many RAM chips are required?
 4. What will be the size of address decoder?

1. What is the memory capacity of the processor?

Capacity = Address Space * Word size

Address lines * Word size

$$= 2^{16} \times 8 \text{ bits}$$

$$= 2^{16} \text{ B}$$

$$= 64 \text{ KB}$$

2. How many address lines are required for RAM?

Address space = $2^{\text{Address lines}}$

$\Rightarrow 4096 = 2^{\text{A.L.}}$

$\Rightarrow 2^{12} = 2^{\text{A.L.}}$

$\therefore \text{Address Lines} = 12$

3. How many RAM chips are required?

$$\frac{2^{16} \times 8}{2^{12} \times 8} = 2^4$$

$$\frac{2^{16} \times 8}{2^{12} \times 8} = 2^4 = 16$$

4. What will be the size of address decoder?

4 : 16

Input → Output

$2^4 = 16$

16 ram chips are required, so 4+16 line decoder needed to interface all 16 ram chips.

Example: 02

- Interface two chips of 4k RAM and 1 chip of 2k ROM. Starting address of RAM is 8000H.

Assuming the processor is 16 bit.

End address = $8000H + \frac{(2^{10} \times 4 - 1)}{16}$

$4K \times 8 = 2^{12}$

RAM1 = 81 0 0 0 | 0 0 0 0 | 0 0 0 0 | 0 0 0 0 = 8000H

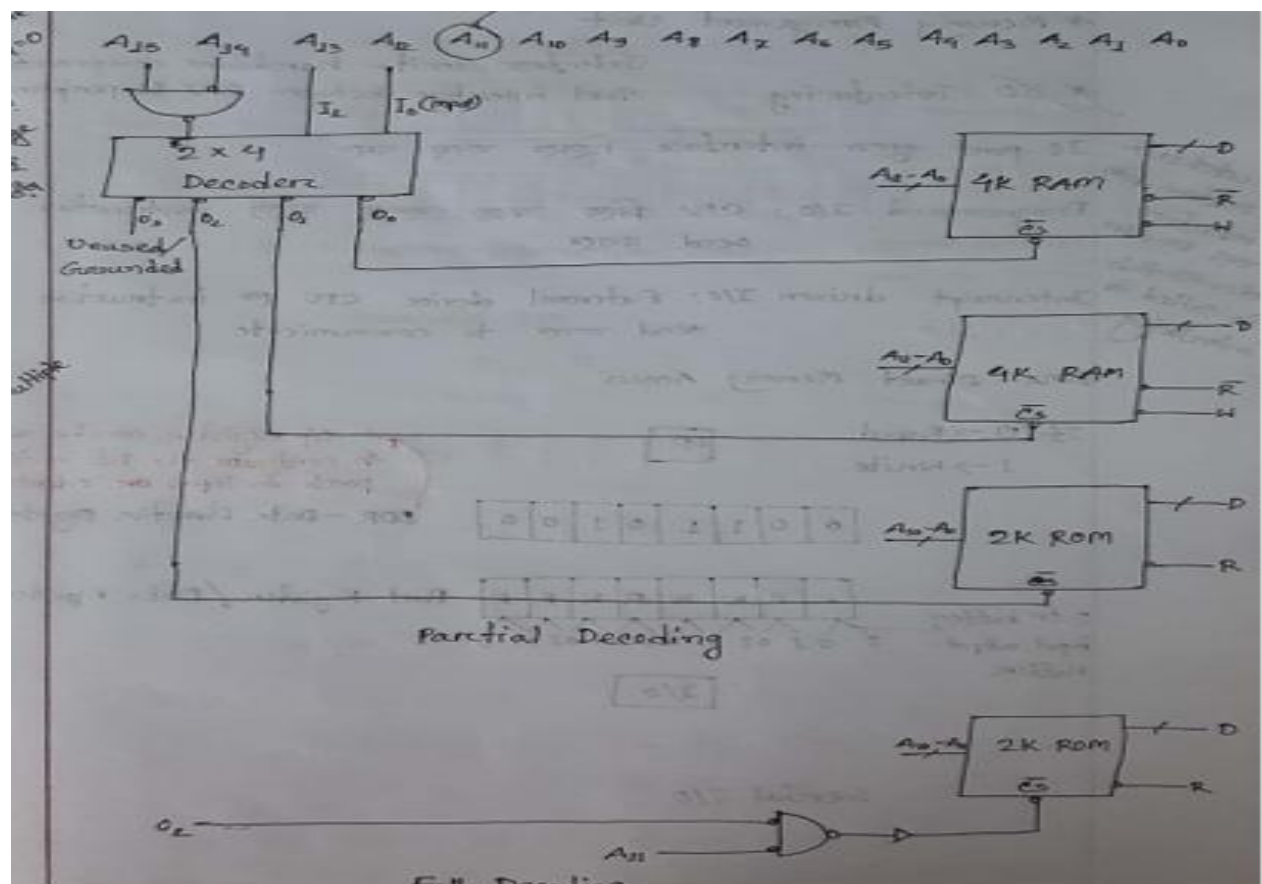
RAM1 = 1 0 0 0 | 1 1 1 1 | 1 1 1 1 | 1 1 1 1 = 8FFFH

RAM2 = 1 0 0 1 | 0 0 0 0 | 0 0 0 0 | 0 0 0 0 = 9000H

RAM2 = 1 0 0 1 | 1 1 1 1 | 1 1 1 1 | 1 1 1 1 = 9FFFH

ROM = 1 0 1 0 | X 0 0 0 | 0 0 0 0 | 0 0 0 0 = A000H - AF

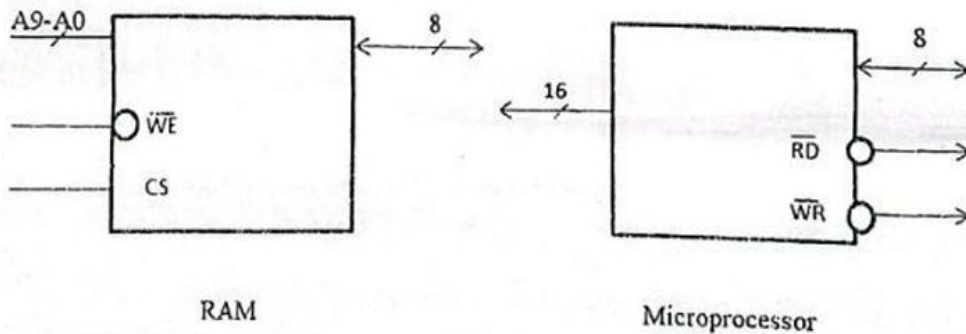
ROM = 1 0 1 0 | X 1 1 1 | 1 1 1 1 | 1 1 1 1 = AFFFH



8279 slide
Interface two chips of 4k RAM and two chips of 2k ROM with processor. Starting address of RAM is A000H. Draw a neat logic diagram of your implementation.

(a). Assume the following RAM chip and the microprocessor:

6



Connect the microprocessor to the RAM to obtain the memory map: 0000_{16} through $0FFF_{16}$. Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

(a) From the given memory

Processor Address bus = 16 bit
Data bus = 8 bit

For RAM, - Address bus = 10 bit
Data bus = 8 bit

Memory map : 0000_{16} to $0FFF_{16}$

∴ So the decoding technique is : Full decoding.

$$\text{Processor capacity} = 2^{16} \times 8 \\ = 64 \text{KB}$$

$$\text{RAM capacity} = 2^{10} \times 8 \\ = 1 \text{KB}$$

$$\text{From memory map} = 0FFF - 0000$$

$$= 0FFF$$

$$= 0 \times 16 + 15 \times 16^2 + 15 \times 16^1 + 15 \times 16^0$$

$$= 4096 \text{ locations}$$

$$\text{Total memory locations} = 4096$$

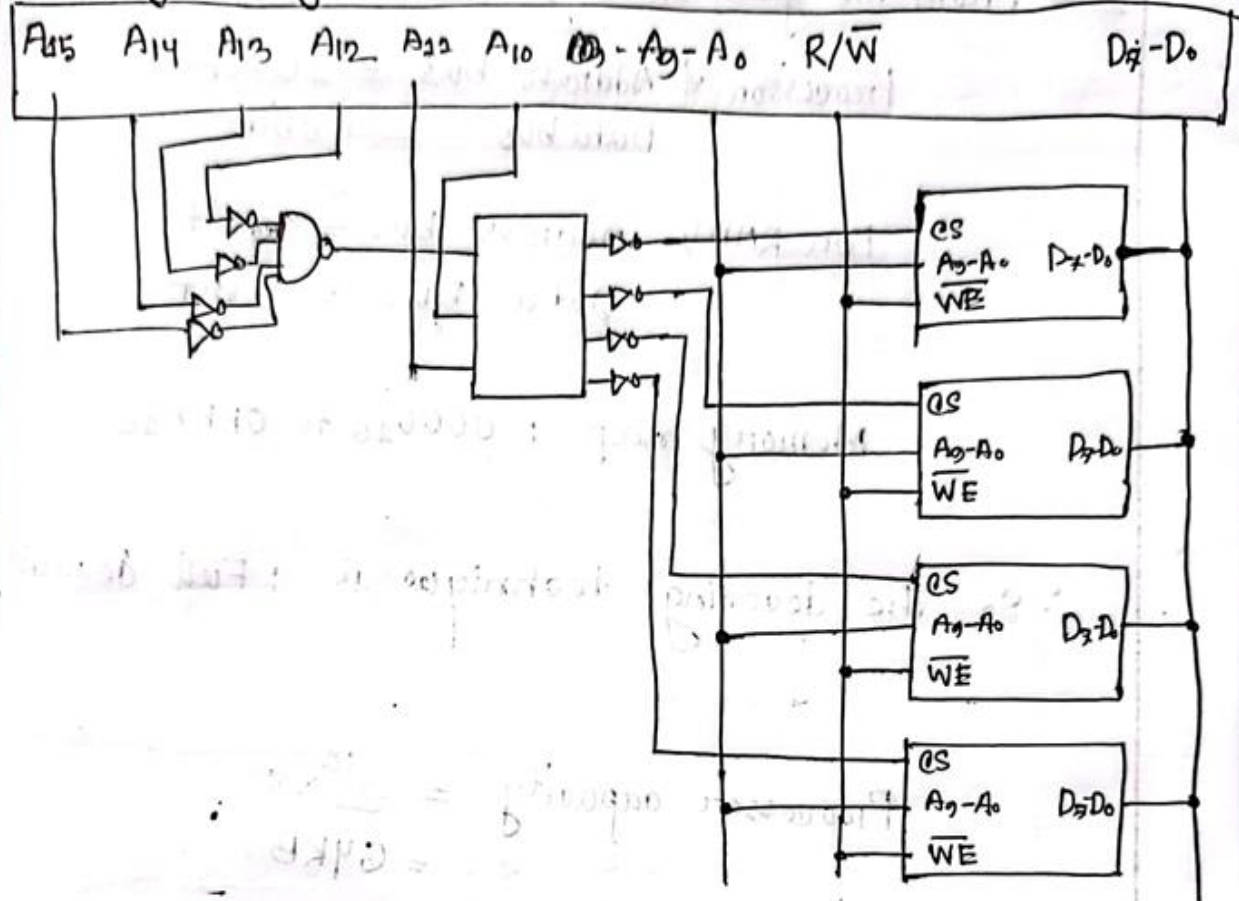
$$\text{Each RAM has} = 1024$$

∴ We need 4 ~~RAM~~ 1KB RAM

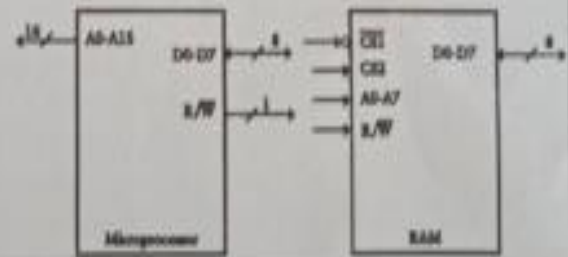
The memory mapping:

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	RAM CHIP 1	0000H to 03FFH
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	RAM CHIP 2	0400H to 07FFH
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	RAM CHIP 3	0800H to 0BFFH
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	RAM CHIP 4	0C00H to 0FFFH

The logic diagram for circuit



Assume the following microprocessor and the RAM chip:
 Draw a neat logic diagram showing connections between the above microprocessor and the RAM chip using only the signals shown to include the memory map (020000). Use linear decoding.



Given that

8-bit microprocessor : Address bus = 16 bit

$$\therefore \text{Processor capacity} = 2^{16} \times 8$$

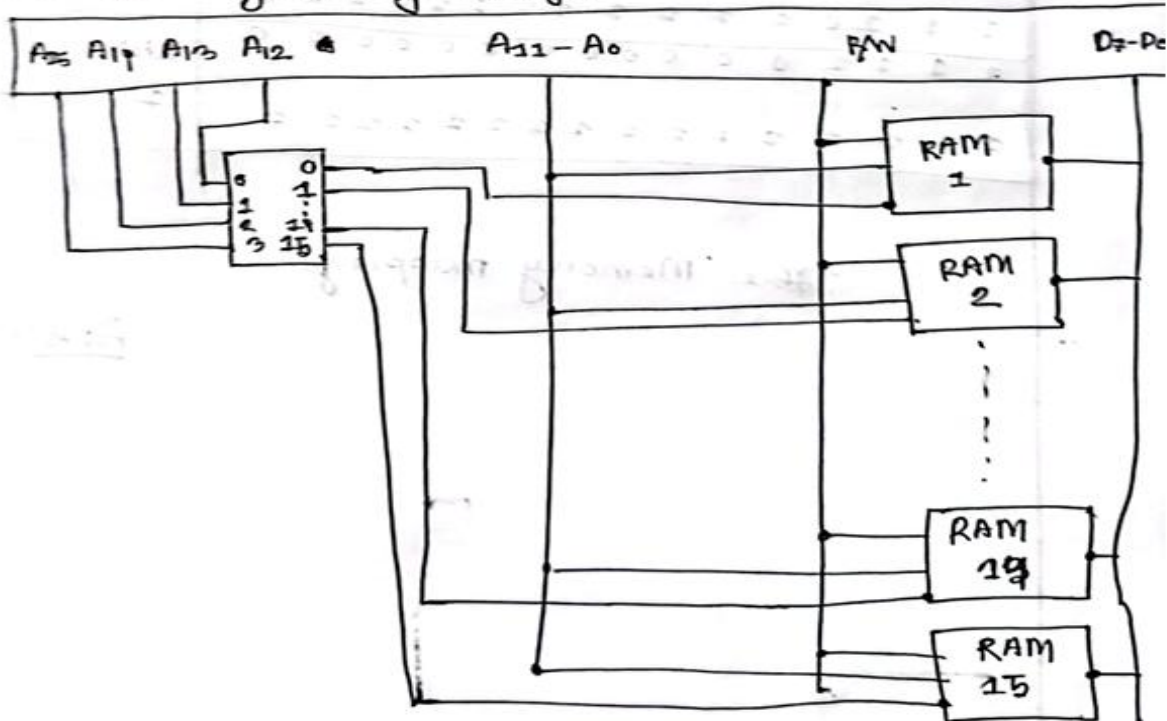
$$= 64 \text{ KB}$$

RAM capacity = 4 KB

$\therefore$ RAM Address line = 12 bit

$$\therefore \text{Number of RAM chip Needed} = \frac{64}{4} = 16 \text{ RAM}$$

For memory Array Design



Address memory mapping																		
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	RAM chip	Address	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	RAM 0	0000H 0FFFH	
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1			
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	RAM 1	1000H 1FFFH	
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1			
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	RAM 14	E000H EFFFH	
1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1			
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	RAM 15	F000H FFFFH	
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1			

4.1 Define the three types of I/O. Identify each as either microprocessor-initiated or device-initiated

Type of I/O	Definition	Description	Initiated By
Programmed I/O (Polling)	CPU-controlled I/O method where processor manages data transfer	CPU continuously checks device status	Microprocessor-initiated
Interrupt-driven I/O	I/O method where device interrupts CPU when ready	CPU services device only when interrupt occurs	Device-initiated
Direct Memory Access (DMA)	I/O method where data transfers directly between device and memory	CPU not involved during transfer	Device-initiated

4.4 What is the difference between memory mapping in a microprocessor and memory-mapped I/O?

Feature	Memory Mapping (Normal Memory)	Memory-Mapped I/O	
Definition	Assigning addresses to RAM/ROM	Assigning memory addresses to I/O devices	
Purpose	Used for memory storage	Used for communication with devices	
Address Space	Separate for memory	Shared with I/O	
Instructions	Memory instructions	Same memory instructions	
Effect	Full memory available	Reduces available memory	

4.8 Summarize the basic difference between maskable and nonmaskable interrupts.

Describe how power failure interrupt is normally handled.

Feature	Maskable Interrupt	Non-Maskable Interrupt	
Definition	Interrupt that can be enabled or disabled by software	Interrupt that cannot be disabled	
Control	Can be masked	Cannot be masked	
Priority	Lower	Highest	
Usage	Normal I/O operations	Critical situations	
Example	Keyboard interrupt	Power failure	

4.10 Discuss the basic difference between internal and external interrupts

Feature	Internal Interrupt	External Interrupt
Definition	Interrupt generated inside CPU due to program execution	Interrupt generated by external hardware devices
Source	Inside CPU	Outside CPU
Cause	Program errors (overflow, divide by zero)	Hardware signals
Example	Arithmetic error	Keyboard, I/O device

Chapter-3

3.1 What is the basic difference between main memory and secondary memory?

Feature	Main Memory	Secondary Memory
Definition	Memory directly accessed by CPU during execution	Permanent storage used to store data long-term
Volatility	Mostly volatile (data lost when power off)	Non-volatile (data remains after power off)
Speed	Very fast	Slower compared to main memory
Capacity	Smaller	Very large
Cost	Expensive per bit	Cheaper per bit
Examples	RAM	Hard disk, SSD, flash drive

3.2 A microprocessor has 24 address pins. What is the maximum size of the main memory?

A microprocessor with 24 address pins can form 2^{24} distinct addresses. If each address selects one byte (typical byte-addressable memory):

- Number of addressable bytes = $2^{24} = 16,777,216$ bytes
- In more convenient units:
 - = 16,777,216 B
 - = 16,384 KB = 16 MB (where 1 KB = 1024 B, 1 MB = 1024 KB)

Therefore the maximum main memory size = 16 MB (16,777,216 bytes).

3.3 Can the microprocessor execute programs directly in hard disk? Explain your answer

A microprocessor cannot execute programs directly from the hard disk **because** it can only access instructions from **main memory (RAM)**, not directly from secondary storage. Therefore, programs must first be loaded into RAM before execution.

3.5 Given a memory with a 14-bit address and an 8-bit word size.

- (a) How many bytes can be stored in this memory?
- (b) If this memory were constructed from $1K \times 1$ RAMs, how many memory chips would be required?
- (c) How many bits would be used for chip select?

(a) How many bytes can be stored?

Number of addressable locations:

$$2^{14} = 16384 \text{ locations}$$

Each location = 1 byte (8-bit word)

So total memory:

$$16384 \text{ bytes} = 16 \text{ KB}$$

(b) Number of $1K \times 1$ RAM chips required

Each chip is $1K \times 1$, meaning:

- $1K = 1024$ locations
- Each location = 1 bit

Step 1: Total memory needed in bits

$$16384 \text{ bytes} = 16384 \times 8 = 131072 \text{ bits}$$

Step 2: Bits per chip

$$1K \times 1 = 1024 \text{ bits per chip}$$

Step 3: Number of chips

$$\frac{131072}{1024} = 128 \text{ chips}$$

(c) How many bits used for chip select?

We need to find how many groups (chips per address space arrangement).

Each chip stores:

- $1K = 2^{10}$ addresses

Total memory addresses:

- 2^{14}

So remaining address bits for chip selection:

$$14 - 10 = 4 \text{ bits}$$

3.8

- (a) How many address and data lines are required for a $1M \times 16$ memory chip?
- (b) What is the size of a decoder with one chip enable ($\overline{CE}$) to obtain a $64K \times 32$ memory from $4K \times 8$ chips? Where are the inputs and outputs of the decoder connected?

(a)

Given:

- Memory size = $1\text{M} \times 16$
- $1\text{M} = 2^{20}$ locations
- Each location = 16 bits

Address lines:

$$2^{20} \Rightarrow 20 \text{ address lines}$$

Data lines:

- Word size = 16 bits
→ 16 data lines



(b)

Given:

- Memory required = $64\text{K} \times 32$
- Chip available = $4\text{K} \times 8$

For memory depth (64K from 4K)

$$64\text{K} / 4\text{K} = 16$$



So we need 16 groups

For data width (32-bit from 8-bit)

$$32/8 = 4$$

So each group needs 4 chips in parallel

Total chips:

$$16 \times 4 = 64 \text{ chips}$$

We have:

- 16 groups

So decoder must select **16 outputs**

$$16 = 2^4$$

✓ So decoder size = 4-to-16 decoder

Inputs of decoder:

- Higher address bits: A12 – A15 (4 bits)

Outputs of decoder:

- Go to Chip Enable (E) pins of memory group

3.9 A microprocessor with 24 address pins and eight data pins is connected to a $1K \times 8$ memory chip with one chip enable. How many unused address bits of the microprocessor are available for interfacing other $1K \times 8$ memory chips? What is the maximum directly addressable memory available with this microprocessor?

(1) Unused address bits for other $1K \times 8$ chips

Step 1: Address lines used by one 1K chip

- $1K = 2^{10}$
→ so 10 address lines are used

Step 2: Total address lines

- Microprocessor has 24 address lines

Step 3: Unused address bits

$$24 - 10 = 14$$

(2) Maximum directly addressable memory

With 24 address lines:

$$2^{24} \text{ locations}$$

Each location = **1 byte** (since 8 data pins = 8-bit word = 1 byte)

Step 1:

$$2^{24} = 16,777,216 \text{ bytes}$$

Step 2:

Convert to MB:

$$16,777,216 = 16 \text{ MB}$$

3.10 Name the methods used in main memory array design. What are the advantages and disadvantages of each?

2) Partial Decoding

Advantages	Disadvantages
Moderate design complexity	Possible address overlap (aliasing)
Less hardware than full decoding	Not fully efficient memory use
Faster than full decoding	Some memory locations may repeat
Common in medium systems	Not completely precise addressing

3) Full Decoding

Advantages	Disadvantages
Full and accurate memory usage	Complex design
No address overlap	Requires more hardware (decoders)
Efficient and reliable	Higher cost
Suitable for large systems	Slightly slower due to decoding

3.7 Explain briefly the differences between segmentation and paging

Feature	Segmentation	Paging
Basic idea	Memory is divided into variable-sized segments (like program parts)	Memory is divided into fixed-size blocks (pages)
Size basis	Variable size (depends on program structure)	Fixed size pages/frames
Programmer view	Visible to programmer (logical view of program)	Invisible to programmer (handled by OS)
Table used	Segment table	Page table
Flexibility	More flexible (logical organization)	Less flexible but simpler to manage

3.19 What basic functions are performed by a microprocessor's on-chip MMU?

Function	Description	
Address Translation	Converts logical/virtual address into physical address used in main memory	
Memory Protection	Prevents a process from accessing unauthorized memory areas	
Paging Support	Helps divide memory into fixed-size pages and manage them	
Segmentation Support	Supports logical division of memory into segments (code, data, stack)	
Access Control	Controls read, write, and execute permissions for memory	
Efficient Memory Use	Helps reduce fragmentation and supports virtual memory system	

- 3.11** The block diagram of a 512×8 RAM chip is shown in Figure P3.11. In this arrangement the memory chip is enabled only when $\overline{CS1} = L$ and $CS2 = H$. Design a $1K \times 8$ RAM system using this chip as the building block. Draw a neat logic diagram of your implementation. Assume that the microprocessor can directly address $64K$ with a $R/\overline{W}$ and eight data pins. Using linear decoding and don't-care conditions as 1's, determine the memory map in hexadecimal.

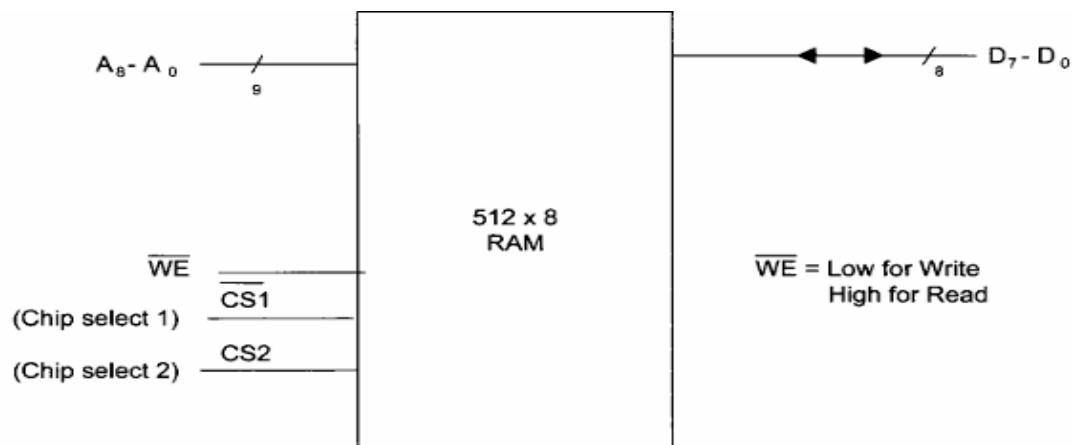
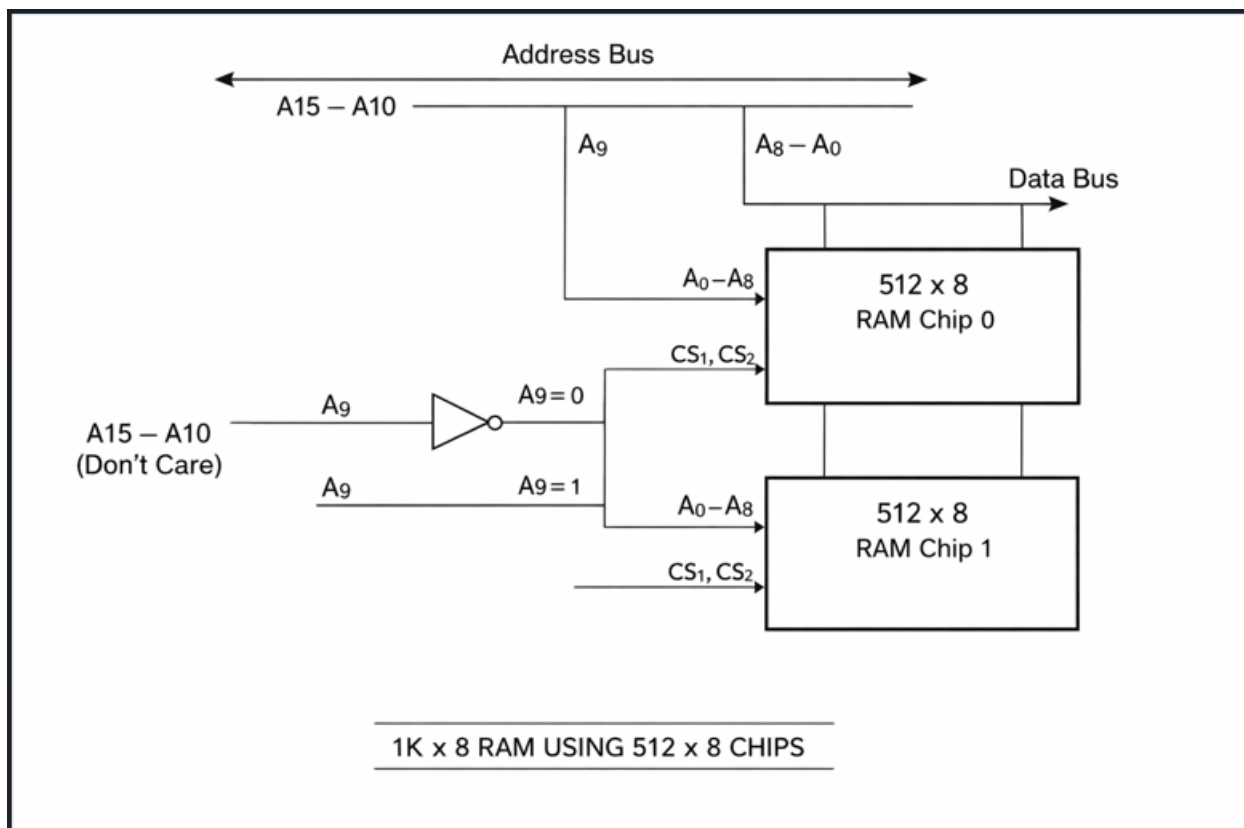


FIGURE P3.11

Answer:



✓ Given

- RAM chip = 512×8
- Required memory = $1K \times 8$
- CPU = 64K address space

✓ Step 1: Number of chips

$$1K/512 = 2 \text{ chips}$$

✓ Step 2: Address lines

- $1K = 2^{10} \rightarrow 10$ address lines
- $512 = 2^9$

So:

- A0–A8 $\rightarrow$ inside chip
- A9 $\rightarrow$ chip select
- A10–A15 $\rightarrow$ don't care

✓ Step 3: Memory map (Hex)

Address Range	Chip
0000H – 01FFH	Chip 0
0200H – 03FFH	Chip 1

3.12 Consider the hardware schematic shown in Figure P3.12.

- Determine the address map of this system. *Note:* $\overline{\text{MEMR}} = 0$ for read, $\overline{\text{MEMR}} = 1$ for write, $\text{M}/\overline{\text{IO}} = 0$ for I/O and $\text{M}/\overline{\text{IO}} = 1$ for memory.
- Is there any possibility of bus conflict in this organization? Clearly justify your answer.

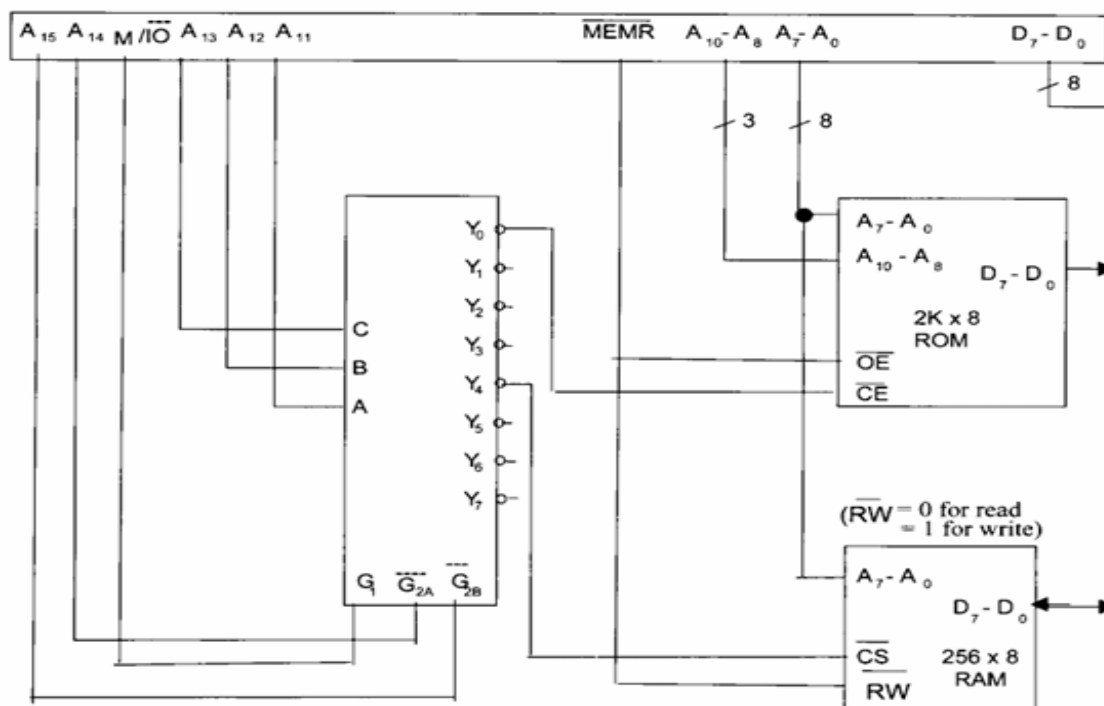


FIGURE P3.12

- 3.13** Interface a microprocessor with 16-bit address pins and 8-bit data pins and a $\overline{R/\overline{W}}$ pin to a $1K \times 8$ EPROM chip and two $1K \times 8$ RAM chips such that the memory map shown in Table P3.13 is obtained:
 Assume that both EPROM and RAM chips contain two enable pins: $\overline{CE}$ and $\overline{OE}$ for the EPROM and $\overline{CE}$ and $\overline{WE}$ for each RAM. Note that $\overline{WE} = 1$ and $\overline{WE} = 0$ indicate read and write operations for the RAM chip. Use a 74138 decoder.

TABLE P3.13

Device	Size	Address Assignment (Hex)
EPROM chip	$1K \times 8$	8000–83FF
RAM chip 0	$1K \times 8$	9000–93FF
RAM chip 1	$1K \times 8$	C000–C3FF

The microcomputer communicates with an external device via one or more registers called I/O ports using programmed I/O. Illustrate the functions of two of them with brief description. 4

- i. What is fluorescence?
- ii. If we use 2-byte pixel values in a 24-bit lookup table representation, how many bytes does the lookup table occupy?

2.15 If we use 2-byte pixel values in a 24-bit lookup table representation, how many bytes does the lookup table occupy?

SOLUTION

$$2^{16} \times 24/8 = 65,536 \times 3 = 196,608.$$

2.16 True or false: fluorescence is the term used to describe the light given off by a phosphor after it has been exposed to an electron beam. Explain your answer.

SOLUTION

False. Phosphorescence is the correct term. Fluorescence refers to the light given off by a phosphor while it is being exposed to an electron beam.

Serialize the steps associated in turning on or off a pixel in LCD monitor.

2.1 What is the resolution of an image?

SOLUTION

The number of pixels (i.e., picture elements) per unit length (e.g., inch) in the horizontal as well as vertical direction.

2.2 Compute the size of a 640×480 image at 240 pixels per inch.

SOLUTION

$640/240$ by $480/240$ or $2\frac{2}{3}$ by 2 inches.

2.3 Compute the resolution of a 2×2 inch image that has 512×512 pixels.

SOLUTION

$512/2$ or 256 pixels per inch.

2.4 What is an image's aspect ratio?

SOLUTION

The ratio of its width to its height, measured in unit length or number of pixels.

2.5 If an image has a height of 2 inches and an aspect ratio of 1.5, what is its width?

SOLUTION

$\text{width} = 1.5 \times \text{height} = 1.5 \times 2 = 3$ inches.

- 2.6 If we want to resize a 1024×768 image to one that is 640 pixels wide with the same aspect ratio, what would be the height of the resized image?

SOLUTION

$$\text{height} = 640 \times 768 / 1024 = 480.$$

- 2.7 If we want to cut a 512×512 sub-image out from the center of an 800×600 image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image?

SOLUTION

$$[(800 - 512)/2, (600 - 512)/2] = (144, 44).$$

- 2.8 Sometimes the pixel at the upper left corner of an image is considered to be at the origin of the pixel coordinate system (a left-handed system). How to convert the coordinates of a pixel at (x, y) in this coordinate system into its coordinates (x', y') in the lower-left-corner-as-origin coordinate system (a right-handed system)?

SOLUTION

$$(x', y') = (x, m - y - 1) \text{ where } m \text{ is the number of pixels in the vertical direction.}$$

- 2.9 Find the CMY coordinates of a color at $(0.2, 1, 0.5)$ in the RGB space.

SOLUTION

$$(1 - 0.2, 1 - 1, 1 - 0.5) = (0.8, 0, 0.5).$$

- 2.10 Find the RGB coordinates of a color at $(0.15, 0.75, 0)$ in the CMY space.

SOLUTION

$$(1 - 0.15, 1 - 0.75, 1 - 0) = (0.85, 0.25, 1).$$

- 2.11 If we use direct coding of RGB values with 2 bits per primary color, how many possible colors do we have for each pixel?

SOLUTION

$$2^3 \times 2^3 \times 2^3 = 4 \times 4 \times 4 = 64.$$

- 2.12** If we use direct coding of RGB values with 10 bits per primary color, how many possible colors do we have for each pixel?

SOLUTION

$$2^{10} \times 2^{10} \times 2^{10} = 1024^3 = 1,073,741,824 > 1 \text{ billion.}$$

- 2.13** The direct coding method is flexible in that it allows the allocation of a different number of bits to each primary color. If we use 5 bits each for red and blue and 6 bits for green for a total of 16 bits per pixel, how many possible simultaneous colors do we have?

SOLUTION

$$2^5 \times 2^5 \times 2^6 = 2^{16} = 65,536.$$

- 2.14** If we use 12-bit pixel values in a lookup table representation, how many entries does the lookup table have?

SOLUTION

$$2^{12} = 4096.$$

- 2.15** If we use 2-byte pixel values in a 24-bit lookup table representation, how many bytes does the lookup table occupy?

SOLUTION

$$2^{16} \times 24/8 = 65,536 \times 3 = 196,608.$$

- 2.16** True or false: fluorescence is the term used to describe the light given off by a phosphor after it has been exposed to an electron beam. Explain your answer.

SOLUTION

False. Phosphorescence is the correct term. Fluorescence refers to the light given off by a phosphor while it is being exposed to an electron beam.

2.17 What is persistence?

SOLUTION

The duration of phosphorescence exhibited by a phosphor.

2.18 What is the function of the control electrode in a CRT?

SOLUTION

Regulate the intensity of the electron beam.

2.19 Name the two methods by which an electron beam can be bent?

SOLUTION

Electrostatic deflection and magnetic deflection.

2.20 What do you call the path the electron beam takes when returning to the left side of the CRT screen?

SOLUTION

Horizontal retrace.

2.21 What do you call the path the electron beam takes at the end of each refresh cycle?

SOLUTION

Vertical retrace.

Types of Video Display Devices Available



Terminology related to display devices

Resolution : Resolution is the number of distinct pixels in each dimension that can be displayed. More the resolution more clear is the image .

Pixel : A pixel or the picture element is the smallest item of information in an image. Graphic monitors display pictures by dividing the display screen into thousands of pixels arranged in rows and columns. Each image is made up of thousands of small pixels.

Aspect ratio of the image is the ratio of the number of X pixels to the number of Y pixels.
The standard aspect ratio for a PC is 4:3.

Resolution	Number of pixels	Aspect Ratio
320 * 200	64,000	8:5

Refreshing : When dot of phosphor is struck by an electron beam it glows for some time and then fades away (persistence). In order to maintain a stable image the electron beam has to redraw the image number of times per second. This is refreshing. Normally in a CRT, the screen is refreshed 60 to 80 frames per second.

additive process

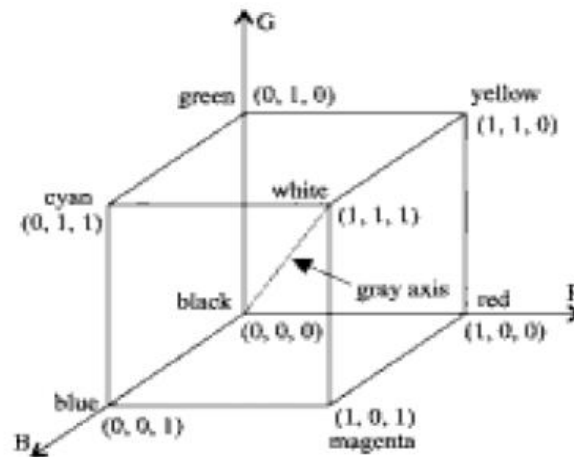


Fig. 2-1 The RGB color space.

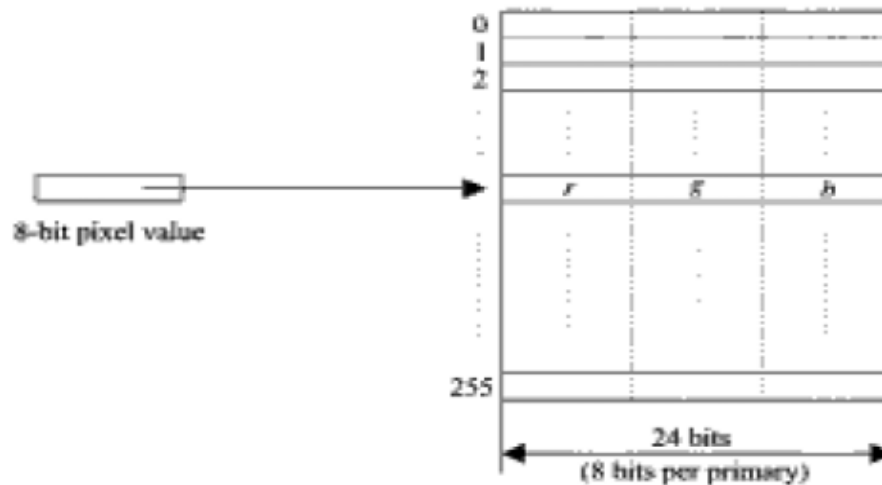
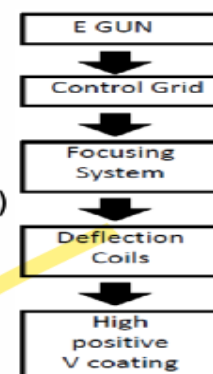


Fig. 2-4 A 24-bit 256-entry lookup table.

Components of CRT

- Produces constant stream of electrons
- Sets intensity of spot on screen (the more negative the control grid voltage the fewer electrons pass through)
- Forces e-beam into narrow stream (otherwise repel)
- Indicates target phosphor spot
- 15,000 - 20,000 V accelerates e-beam to screen



CRT Working Principle

- CRT display works using :
 - **Electron emission**
 - Electrons are emitted from the Cathode tube.
 - **Phosphorescence**
 - It is the emission of visible light, when electron beam strikes Phosphor material.

CRT Monitor Displays color pictures by using a combination of phosphors that emit different colored light.

Two Techniques for producing color Display with a CRT are:-

1. Beam- Penetration
2. Shadow Mask

Beam Penetration Method:

It produces a limited number of colors red, orange, yellow and green. Picture Quality is not very good.

Shadow Mask Method:

It can display a much wider range than Beam Penetration Method. It has three phosphor color dots (red, green and blue) at each pixel position.

CRT Monitor Technology



CRT Display: Advantages and Disadvantages

- **Advantages:**
 - Offers greater resolution .
 - Widest viewing angle when compared to any other technology.
 - It is cheap as compared to LCD, PLASMA displays.
- **Disadvantages:**
 - Thickness is much larger as compared to LCD, PLASMA display.
 - Cannot be used for smaller displays like watches, calculators, portable devices.
 - View area is less than the offered monitor size.
 - It is more fragile and bulky.

There are two categories of flat panel displays:

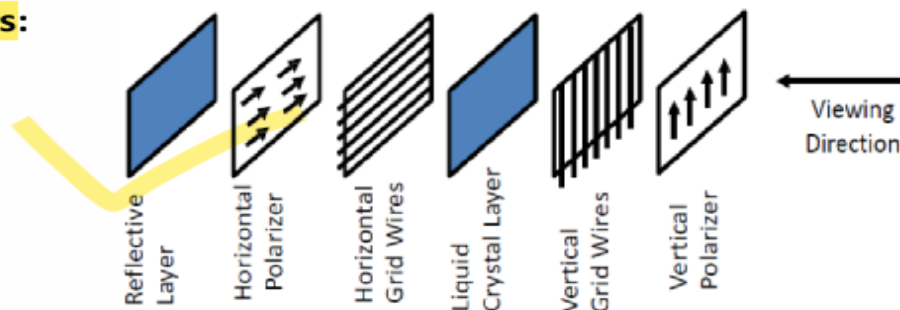
- Emissive Displays
- Non-Emissive Displays

Emissive Displays (or emitters) are devices that convert electrical energy into light. Plasma Panel (PDP) and light emitting diodes (LED) are examples of **emissive displays**.

Non-emissive displays (or non-emitters) use optical effects to convert sunlight or light from some other source into graphics pattern e.g Liquid Crystal Display (LCD)

LCD Construction

- Each pixel of an LCD consists of a layer of molecules aligned between two transparent electrodes, and two polarizing filters, whose axes are perpendicular.
- Orientation of the liquid crystal molecules is determined by the alignment at the surfaces.
- **Six Layers:**



Classification of LCD: – Transmissive displays. Ex: Computer display.

_ Reflective displays. Ex: Calculator.

Applications, Advantages & Limitations of LCD

- **Applications:**

- LCDs with a small number of segments, are used in digital watches and pocket calculators, Small monochrome displays in personal organizers, or older laptop screens
- High-resolution color displays such as modern LCD computer monitors and televisions;

- **Advantages:**

- An LCD cell consumes only microwatts of power over a thousand times less than LED displays.
- LCDs can operate on voltages as low as 2 to 3 V and are easily driven by MOS IC drivers.

- **Limitations:**

- They cannot be seen in the dark,
- They have a limited viewing angle;
- They can be operated in limited temperature range.

Advantages and Disadvantages of TFT

- **Advantages:**

- Fast response time has been a great feature of TN (twisted nematics) displays.
- Lower cost of production
- Less noise or glitter seen on the panel surface

- **Disadvantages:**

- The TFT display suffers from limited viewing angles, especially in the vertical direction.
- They are unable to display the full 16.7 million colors (24-bit true color) available from modern graphics cards.

Plasma Display

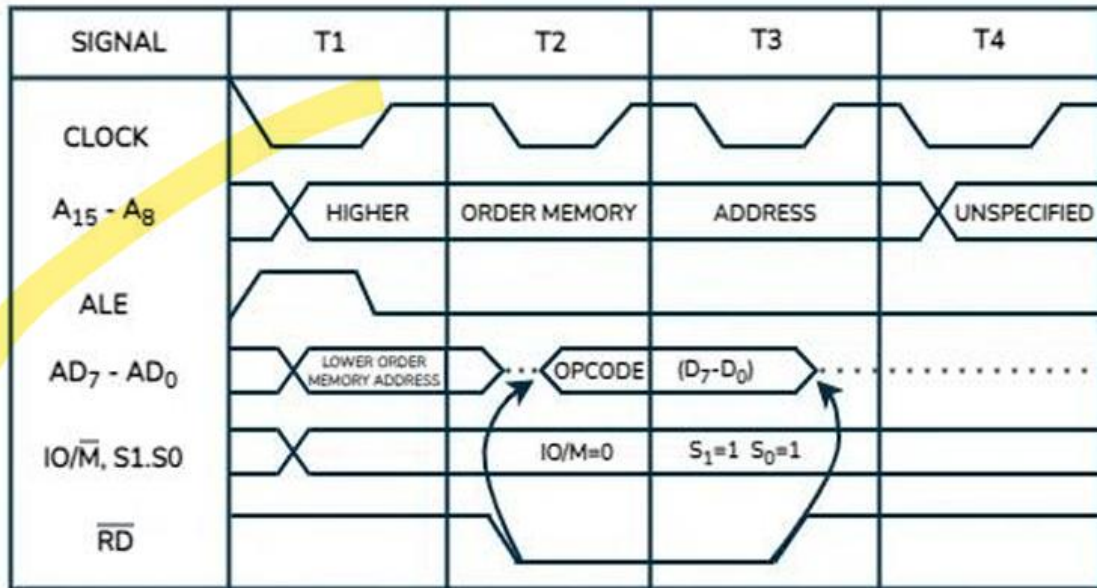
- A **plasma display panel (PDP)** is a type of flat panel display, suitable and now commonly used for large TV displays (typically above 37-inch or 940 mm).

Advantages and Disadvantages

- Advantages:
 - Supports large displays ,up to 103 inches diagonally.
 - Overall thickness of monitor is less than 10 cm. and can be installed on a wall.
 - Faster response time, Greater color spectrum, Wider viewing angle
- Disadvantages:
 - They are costly as compared to CRT, LCD displays.
 - Burn-in problem:
 - Static images can leave a permanent mark on screen. That is why they are not used in computer monitors and video game displays.
- The 8085 microprocessor has 5 basic machine cycles.
 - Opcode Fetch {4T- state}
 - Memory Read {3T- state}
 - Memory Write {3T- state}
 - I/O Read {3T- state}
 - I/O Write {3T- state}

Opcode Fetch Machine Cycle:

Each processor instruction has a one byte opcode. the first, 3T-states are used to load the opcode from memory, and the remaining T-states are used for internal opcodes.



Memory Read Machine Cycle: A memory read machine cycle is executed by the processor to read a data byte from memory. The processor takes 3T states to perform.

Memory Write Machine Cycle: A write-to-memory machine cycle is executed by the processor to write a data byte to memory. The processor takes 3T states to perform.

I/O Read Machine Cycle: It is performed by the processor to read a data byte from an I/O port that is I/O mapped in the system. The processor needs 3T states to execute.

I/O Write Machine Cycle: It is executed by the processor to write a data byte to an I/O port that is I/O mapped in the system. The processor needs 3T states to execute.

Applications Of 8085

- System charts keep track of all changes occurring in the system.
- Timelines provide a more accurate representation of physical activity.
- The timing table of 8085 instructions graphically represents the time required to execute each instruction.
- Timing Diagram is a special form of sequence diagram.
- Timing Diagram keep track of every change that occur in the system.
- The timing diagram is important for planning the clock frequency of the microprocessor.
- The timing diagram helps design memory circuits that meet the required set and hold times during read and write operations.

